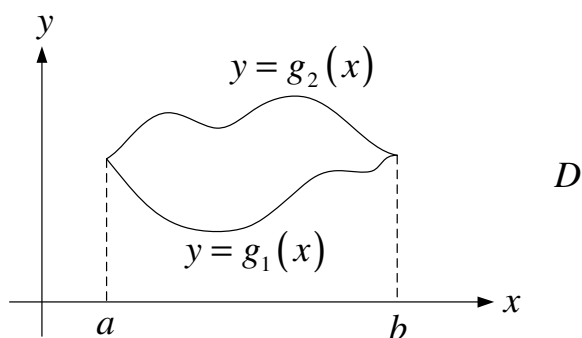


§15.3 Double Integrals over General Regions

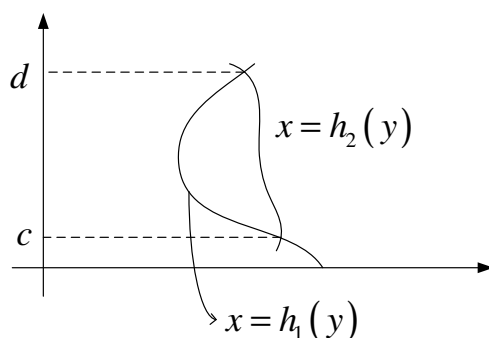
Homework : 3,9,13,17,21,27,28,31,37,41,47,49,51,54,55

General Regions

(1) Tape I



(2) Tape II



- 只要屋頂“不太壞”（如：沒破洞，只有些許裂縫），則 general region 的邊界“不太壞”（如：沒斷點），則一維和二維計算面積方式仍相同 ie。

$$\text{(Tape I)} \quad \iint_D f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx$$

$$\text{(Tape II)} \quad \iint_D f(x, y) dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy$$

Summary :

- (1) D 可能是 (i) Tape I (ii) Tape II (iii) 是 Tape I 也是 Tape II (iv) 二者皆不是但可表成兩種聯集.

$$(2) \text{ 若 } D_1 \cup D_2 = D, D_1 \cap D_2 = \text{邊界} \Rightarrow \iint_D f dA = \iint_{D_1} f dA + \iint_{D_2} f dA.$$

$$(3) \text{ linear : } \iint_D (\alpha f + g) dA = \alpha \iint_D f dA + \iint_D g dA.$$

$$(4) \text{ If } m \leq f(x, y) \leq M, \Rightarrow mA(D) \leq \iint_D f(x, y) dA \leq MA(D).$$

Here $A(D)$ = 地基的面積.

$$(5) \iint_D f(x) dA = 0 \text{ Provided that}$$

(i) D 對稱 y 軸 (利用對稱性) (ii) $f(x)$: odd functions.

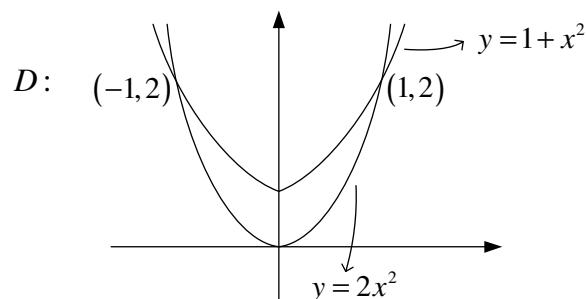
$$\textbf{Example 1 : } \iint_D (x^2 \tan x + y^3 + 4) dA \quad D: x^2 + y^2 = 2$$

Solution :

$$\begin{aligned} & \iint_D (x^2 \tan x + y^3 + 4) dA \\ &= \iint_D x^2 \tan x dA + \iint_D y^3 dA + \iint_D 4 dA \\ &= 0 + 0 + 8\pi \\ &= 8\pi \end{aligned}$$

$$\textbf{Example 2 : } \iint_D (x + 2y) dA, D: \begin{cases} y = 2x^2 \\ y = 1 + x^2 \end{cases}$$

Solution :

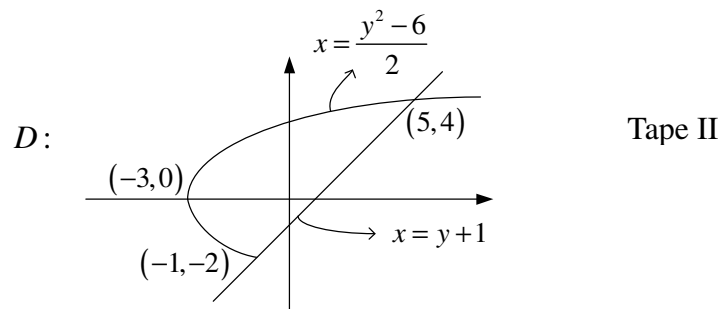


Tape I

$$= \int_{-1}^1 \int_{2x^2}^{1+x^2} (x + 2y) dx = \frac{32}{15}$$

Example 3 : $\iint_D xy dA$, $D: \begin{cases} y = x - 1 \\ y^2 = 2x + 6 \end{cases}$

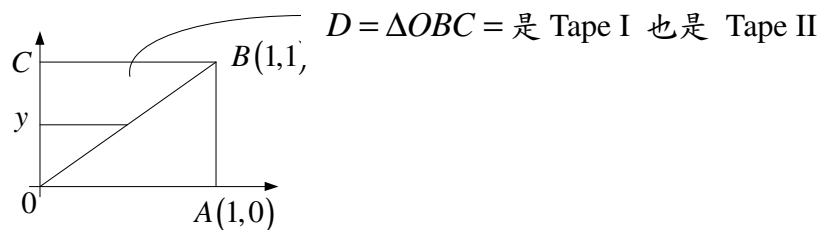
Solution :



$$= \int_{-2}^4 \int_{\frac{y^2}{2}-3}^{y+1} xy dx dy = 36$$

Example 4 : $\int_0^1 \int_x^1 \sin^2 y dy dx$

Solution :

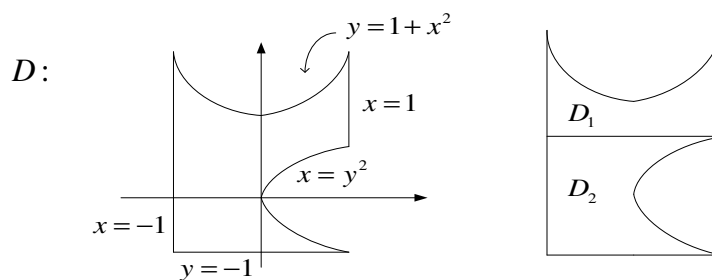


Tape I 寫法不容易積出 $\int_x^1 \sin^2 y dy$ 改成 Tape II 寫法

$$\begin{aligned} &= \int_0^1 \int_0^y \sin^2 x dx dy = \int_0^1 y \sin^2 y dy \\ &= -\frac{\cos y^2}{2} \Big|_0^1 = -\frac{\cos 1}{2} + \frac{1}{2} \end{aligned}$$

Example 5 : $\iint_D xy dA$

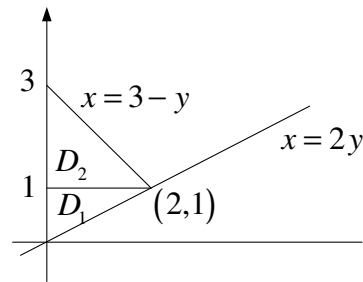
Solution :



$$\begin{aligned}
&= \iint_{D_1} xy dA + \iint_{D_2} xy dA \\
&= \int_{-1}^1 \int_1^{1+x^2} xy dy dx + \int_{-1}^1 \int_{-1}^{y^2} xy dx dy \\
&= 0 \quad (\text{利用對稱來算})
\end{aligned}$$

Example 6 : $\iint_D f(x, y) dA = \int_0^1 \int_0^{2y} f(x, y) dx dy + \int_1^3 \int_0^{3-y} f(x, y) dx dy$. Find D .

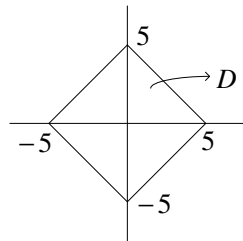
Solution :



$$\Rightarrow D = D_1 \cup D_2$$

Example 7 : $\iint_D (2-3x+4y) dA$

Solution :



(利用對稱性)

$$\begin{aligned}
&\iint_D (2-3x+4y) dA \\
&= \iint_D 2dA - \iint_D 3xdA + \iint_D 4y dA \\
&= 2 \cdot 50 - 0 + 0 \\
&= 50
\end{aligned}$$